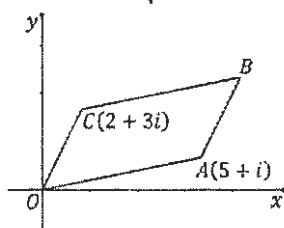


Complex Number 6

- If $z = \cos \theta + i \sin \theta$, show that $z^n + z^{-n} = 2 \cos n\theta$ and that $z^n - z^{-n} = 2i \sin n\theta$. Hence show that
 - $\cos^4 \theta = \frac{1}{8}(\cos 4\theta + 4 \cos 2\theta + 3)$
 - $\sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$
- Factorise $z^3 - 1$. If z is one of the three cube roots of unity, find the two possible values of $z^2 + z + 1$.
- $1, \omega$ and ω^2 are the three cube roots of unity. State the values of ω^3 and $1 + \omega + \omega^2$. Hence simplify each of the expressions $(1 + 3\omega + \omega^2)^2$ and $(1 + \omega + 3\omega^2)^2$ and show that their sum is -4 and their product is 16 .
- Use De Moivre's theorem to find in modulus/argument form
 - The square roots of $\sqrt{3} + i$
 - The cube roots of $-2 - 2i$
- Sketch on an Argand diagram the locus of the point P representing z , given that $|z|^2 = z + \bar{z} + 1$.
- $|z + i| \leq 2$ and $0 \leq \arg(z + 1) \leq \frac{\pi}{4}$. Sketch the region in an Argand diagram which contains the point P representing z .
- $|z - 1| \leq |z - i|$ and $|z - 2 - 2i| \leq 1$. Sketch the region in the Argand diagram which contains the point P representing z . If P describes the boundary of this region, find the value of z when $\arg(z - 1) = \frac{\pi}{4}$.
- $\arg(z + 3) = \frac{\pi}{3}$. Sketch the locus of the point P representing z on an Argand diagram. Find the modulus and argument of z when $|z|$ takes its least value. Hence find, in the form $a + ib$, the value of z for which $|z|$ is a minimum.
- $OABC$ is a parallelogram. The point A and C represent $5 + i$ and $2 + 3i$ respectively. Find the complex number represents by:



- The vector OB
 - The vector AC
 - The vector CA
- Write down the general solution of $\tan 4\theta = 1$.
 - Use De Moivre's Theorem to express $\cos 4\theta$ and $\sin 4\theta$ in terms of $\cos \theta$ and $\sin \theta$.
 - Hence show that $\tan 4\theta = \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta}$.
 - Hence or otherwise, find the roots of the equation $x^4 + 4x^3 - 6x^2 - 4x + 1 = 0$.
 - Hence show that $\tan \frac{\pi}{16} + \tan \frac{5\pi}{16} = \tan \frac{3\pi}{16} + \tan \frac{7\pi}{16} - 4$.
 - If $z = \cos \theta + i \sin \theta$, use De Moivre's theorem to show that $z^n + z^{-n} = 2 \cos n\theta$
 - Hence solve $2z^4 + 3z^3 + 5z^2 + 3z + 2 = 0$. (Solve the roots in $a + bi$ form where a, b are real)
 - If w is a non-real cube root of unity
 - Prove $\frac{1}{1+w} + \frac{1}{1+w^2} = 1$.
 - Show $\frac{1+w^n+w^{2n}}{3} = 1$ or 0 . According to whether n is or is not a multiple of 3 .
 - If $|z - \omega| = |z + \omega|$, find the difference between $\arg(z)$ and $\arg(\omega)$. Show all reasoning.
 - If ω represents one of the complex cube roots of unity, show $(1 - \omega^8)(1 - \omega^4)(1 - \omega^2)(1 - \omega) = 9$

15. In an Argand diagram, the points P , Q and R represent the complex numbers z_1 , z_2 and $z_2 + i(z_2 - z_1)$ respectively.
- Show that ΔPQR is right-angled.
 - Find in terms of z_1 and z_2 the complex number represented by the point S such that $PQRS$ is a rectangle.
16. a. i. If $z = \cos \theta + i \sin \theta$, show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$.
- ii. Given that $z + \frac{1}{z} = \sqrt{2}$, find the value of $z^{10} + \frac{1}{z^{10}}$.
- b. Given that 1 , ω and ω^2 are the cube roots of unity, find the value of $(1 + 2\omega + 3\omega^2)(1 + 2\omega^2 + 3\omega)$
- c. i. Solve $\omega^2 = -11 - 60i$ for ω , writing your answer in the form $\omega = x + iy$ where $x, y \in \mathbb{R}$
- ii. Hence or otherwise, solve the equation $z^2 - (1 + 4i)z - (1 - 17i) = 0$.
- d. In the Argand diagram below, vectors a , b , c represent the complex numbers z_1 , z_2 , $z_1 + z_2$ respectively, where $z_1 = \cos \theta + i \sin \theta$ and $z_1 + z_2 = (1 + i)z_1$.
- Express z_2 in terms of z_1 , and show that $OACB$ is a square.
 - Show that $(z_1 + z_2)(\overline{z_1 - z_2}) = 2i$.
17. Let $z = x + iy$, where $x, y \in \mathbb{R}$ be the complex number satisfying the inequality $z\bar{z} + (1 - 2i)z + (1 + 2i)\bar{z} \leq 4$. Sketch the locus of z on an Argand diagram.
18. Let ω be one of the non-real cube roots of 1.
- Show that $1 + \omega + \omega^2 = 0$
 - Evaluate $(-1 + \omega + \omega^2)(1 - \omega + \omega^2)(1 + \omega - \omega^2)$
 - Evaluate $(1 + \omega)(1 + \omega^2)(1 + \omega^5)(1 + \omega^8)(1 + \omega^{11})$
19. It can be shown $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$ and $\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$.
- Deduce that $\tan 3\theta = \frac{\tan^3 \theta - 3 \tan \theta}{3 \tan^2 \theta - 1}$
 - Hence find the roots of $x^3 - 3x^2 - 3x + 1 = 0$
 - Without using a calculator, evaluate $\tan \frac{\pi}{12} + \tan \frac{5\pi}{12}$.

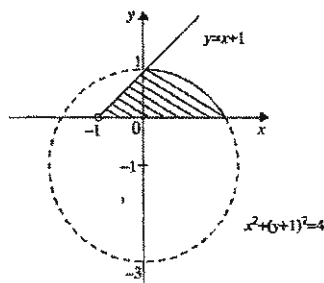
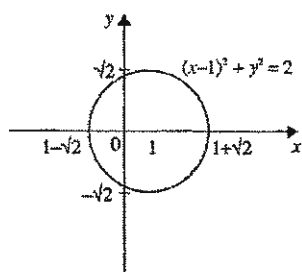
Answer:

2. 0,3

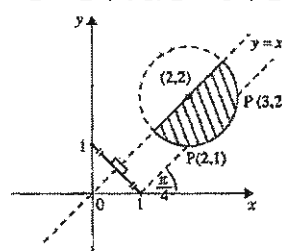
4. $\pm \sqrt{2} \operatorname{cis} \left(\frac{\pi}{12} \right); \sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4} \right), \sqrt{2} \operatorname{cis} \left(\frac{5\pi}{12} \right), \sqrt{2} \operatorname{cis} \left(-\frac{11\pi}{12} \right)$

5.

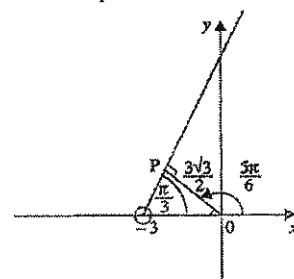
6.



7. $\arg(z-1) = \frac{\pi}{4}$ for
 $z = 2 + i$ or $z = 3 + 2i$



8. $z = \frac{3}{4}(-3 + \sqrt{3}i)$



9. a. $7 + 4i$ b. $-3 + 2i$ c. $3 - 2i$

10. a. $\theta = \frac{n\pi}{4} + \frac{\pi}{16}, n \in \mathbb{Z}$ b.

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta, \sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta.$$

$$x = \tan \frac{\pi}{16}, \tan \frac{5\pi}{16}, -\tan \frac{7\pi}{16}, -\tan \frac{3\pi}{16}$$

11. b. $z = -\frac{1}{4} \pm \frac{\sqrt{15}}{4}i, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$

13. $\pm \frac{\pi}{2}$

15. b. $S = z_1 + i(z_2 - z_1)$

16. a. ii. 0 b. 3 c. i. $\omega = \pm(5 - 6i)$ ii. $z = -2 + 5i, 3 - i$

d. i. $z_2 = iz_1$

17. $(x+1)^2 + (y+2)^2 = 3^2$

18. b. -8 c. -1

19. b. $x = \tan \frac{\pi}{12}, \tan \frac{5\pi}{12}, \tan \frac{3\pi}{4} = -1$ c. 4